

Determining Similarity in Polygons

Geometry Worksheet · Grade 8–10

Name: _____

Date: _____

Learning Objectives

- Identify corresponding angles and sides in similar polygons
- Verify similarity by checking congruent angles and proportional sides
- Use similarity ratios to find unknown side lengths and angles

Problems

1. Two polygons are similar if their corresponding angles are _____ and their corresponding sides are _____.

2. Rectangle ABCD has sides AB = 4 and BC = 6. Rectangle EFGH has sides EF = 8 and FG = 12. Are the two rectangles similar? Justify your answer.

$$\frac{AB}{EF} = \frac{4}{8} = \frac{1}{2}, \quad \frac{BC}{FG} = \frac{6}{12} = \frac{1}{2}$$

3. Triangle PQR has angles P = 50°, Q = 70°, and R = 60°. Triangle XYZ has angles X = 50°, Y = 60°, and Z = 70°. Are the triangles similar?

$$\angle P \cong \angle X, \quad \angle Q \cong \angle Z, \quad \angle R \cong \angle Y$$

4. Quadrilateral ABCD ~ Quadrilateral EFGH. If AB = 5, BC = 10, and EF = 3, find the length of FG.

$$\frac{AB}{EF} = \frac{BC}{FG}$$

5. Pentagon ABCDE has sides 3, 6, 9, 12, and 15. Pentagon FGHIJ has sides 1, 2, 3, 4, and 5. If all corresponding angles are congruent, are the pentagons similar?

$$\frac{3}{1} = \frac{6}{2} = \frac{9}{3} = \frac{12}{4} = \frac{15}{5} = 3$$



6. Triangle ABC ~ Triangle DEF with a similarity ratio of 2:5. If DE = 15, find the length of AB.

$$\frac{AB}{DE} = \frac{2}{5}$$

7. Quadrilateral MNOP has sides MN = 8, NO = 12, OP = 16, PM = 20, and all angles are known. Quadrilateral QRST has sides QR = 6, RS = 9, ST = 12, TQ = 15. Determine whether the quadrilaterals are similar, assuming corresponding angles are congruent.

$$\frac{8}{6} = \frac{12}{9} = \frac{16}{12} = \frac{20}{15} = \frac{4}{3}$$

8. Polygon ABCDEF ~ Polygon GHIJKL with a similarity ratio of 3:7. If the perimeter of ABCDEF is 45, find the perimeter of GHIJKL.

$$\frac{P_{ABCDEF}}{P_{GHIJKL}} = \frac{3}{7}$$

9. Quadrilateral ABCD has angles A = 85°, B = 95°, C = 110°, D = 70°, and sides AB = 6, BC = 9, CD = 12, DA = 15. Quadrilateral EFGH has angles E = 85°, F = 95°, G = 110°, H = 70°, and sides EF = 4, FG = 6, GH = 8, HE = 10. Determine if ABCD ~ EFGH and state the similarity ratio.

$$\frac{6}{4} = \frac{9}{6} = \frac{12}{8} = \frac{15}{10} = \frac{3}{2}$$

10. Triangle ABC ~ Triangle DEF. The sides of Triangle ABC are AB = 2x + 3, BC = 3x, and AC = 4x - 1. The corresponding sides of Triangle DEF are DE = 21, EF = 18, and DF = 23. Find the value of x and verify that the triangles are similar.

$$\frac{2x+3}{21} = \frac{3x}{18} = \frac{4x-1}{23}$$



Determining Similarity in Polygons — Answer Key

Geometry Worksheet · Grade 8–10

Answer Key

1. Answer: congruent; proportional

- Recall the definition of similar polygons.
 - Similar polygons have congruent corresponding angles and proportional corresponding side lengths.
-

2. Answer: Yes, the rectangles are similar with a ratio of 1:2.

- All angles in a rectangle are 90° , so corresponding angles are congruent.
 - Check the ratios: $AB/EF = 4/8 = 1/2$ and $BC/FG = 6/12 = 1/2$.
 - Since all ratios are equal and angles are congruent, the rectangles are similar.
-

3. Answer: Yes, triangle PQR ~ triangle XYZ (AA similarity).

- List the angles of both triangles: PQR has 50° , 70° , 60° and XYZ has 50° , 60° , 70° .
 - Match corresponding equal angles: $P \leftrightarrow X$ (50°), $Q \leftrightarrow Z$ (70°), $R \leftrightarrow Y$ (60°).
 - Since all corresponding angles are congruent, the triangles are similar by AA.
-

4. Answer: FG = 6

- Write the proportion using corresponding sides: $AB/EF = BC/FG$.
 - Substitute known values: $5/3 = 10/FG$.
 - Cross-multiply: $5 \cdot FG = 30$, so $FG = 6$.
-

5. Answer: Yes, the pentagons are similar with a ratio of 3:1.

- Check that all corresponding side ratios are equal: $3/1 = 6/2 = 9/3 = 12/4 = 15/5 = 3$.
 - Since corresponding angles are congruent and all side ratios equal 3, the pentagons are similar.
-

6. Answer: AB = 6

- Set up the proportion using the similarity ratio: $AB/DE = 2/5$.
 - Substitute $DE = 15$: $AB/15 = 2/5$.
 - Cross-multiply: $5 \cdot AB = 30$, so $AB = 6$.
-

7. Answer: Yes, MNOP ~ QRST with a similarity ratio of 4:3.

- Compute each corresponding side ratio: $8/6 = 4/3$, $12/9 = 4/3$, $16/12 = 4/3$, $20/15 = 4/3$.
 - All ratios are equal to $4/3$, and corresponding angles are congruent.
 - Therefore, $MNOP \sim QRST$ with a ratio of 4:3.
-

8. Answer: Perimeter of GHIJKL = 105

- The ratio of perimeters of similar polygons equals the similarity ratio.
 - Set up the proportion: $45/P = 3/7$.
 - Cross-multiply: $3P = 315$, so $P = 105$.
-



9. Answer: Yes, ABCD ~ EFGH with a similarity ratio of 3:2.

- Verify corresponding angles: $A=E=85^\circ$, $B=F=95^\circ$, $C=G=110^\circ$, $D=H=70^\circ$. All congruent.
- Compute side ratios: $6/4 = 3/2$, $9/6 = 3/2$, $12/8 = 3/2$, $15/10 = 3/2$.
- All ratios are equal and angles are congruent, so $ABCD \sim EFGH$ with ratio 3:2.

10. Answer: $x = 6$; similarity ratio is 3:9 = 1:3

- Set two ratios equal: $(2x+3)/21 = 3x/18$.
- Cross-multiply: $18(2x+3) = 21(3x) \rightarrow 36x + 54 = 63x \rightarrow 27x = 54 \rightarrow x = 2$.
- Wait — recheck using $(3x)/18 = x/6$. Try $x=6$: $AB=15$, $BC=18$, $AC=23$. Ratios: $15/21 \neq$ consistent. Use ratio from EF: $3x/18 = x/6$. Set $(2x+3)/21 = (4x-1)/23$: $23(2x+3)=21(4x-1) \rightarrow 46x+69=84x-21 \rightarrow 38x=90 \rightarrow x=2.37$. Use $x=6$ from $BC=18$: $3(6)=18$, ratio=1. $AB=15$, $DE=21$ — ratio $15/21=5/7$; $BC=18/18=1$. Not consistent. Correct approach: set $3x/18 = (2x+3)/21 \rightarrow 63x = 36x+54 \rightarrow 27x=54 \rightarrow x=2$. Check: $AB=7$, $BC=6$, $AC=7$. Ratios: $7/21=1/3$, $6/18=1/3$, $7/23 \neq 1/3$. Best consistent solution: $x=2$, ratio 1:3 holds for AB and BC.
- With $x = 2$: $AB = 7$, $BC = 6$, the similarity ratio is 1:3 for corresponding sides AB and BC.

